

Assignment - 02

Derivation of the Output Voltage Equation for Class A, B, C, D, and E Choppers

Objective

This assignment derives, from first principles, the average output-voltage equation of each of the five standard chopper (DC-DC converter) classes — A, B, C, D, and E. For every class the derivation is presented step by step, followed by a physical explanation of why the resulting equation makes that chopper a step-up or step-down (or both) converter. Each derivation is supported by a Falstad circuit simulation (where available), the corresponding switching / output waveform captured from that simulation, and a plot of the output voltage as a function of duty cycle D .

1. Class A Chopper (First-Quadrant / Step-Down Chopper)

Circuit Description:

A Class A chopper uses a single controlled switch in series with the source and a freewheeling diode across the load. It allows power flow in only one direction (source to load) and produces only a positive output voltage and positive output current — hence "first-quadrant" operation. It is the chopper equivalent of a Buck (step-down) converter.

Step-by-Step Derivation:

Step 1: Define the switching period T , which consists of an ON interval T_{on} (switch closed) and an OFF interval $T_{off} = T - T_{on}$ (switch open).

Step 2: During T_{on} , the switch connects the load directly across the source, so the instantaneous terminal voltage equals the source voltage: $v(t) = V_s$.

Step 3: During T_{off} , the switch opens and the inductive load current is diverted through the freewheeling diode, which effectively shorts the output terminals: $v(t) = 0$.

Step 4: The average (DC) output voltage over one period is found by integrating the instantaneous voltage over one full cycle and dividing by the period: $V_o = (1/T) \int_0^T v(t) dt$.

Step 5: Split the integral at $t = T_{on}$, using $v(t) = V_s$ for $0 \leq t \leq T_{on}$ and $v(t) = 0$ for $T_{on} \leq t \leq T$:

$$V_o = (1/T) \left[\int_0^{T_{on}} V_s dt + \int_{T_{on}}^T 0 dt \right]$$

Step 6: Evaluate the first integral (the second is zero):

$$V_o = (V_s \cdot T_{on}) / T$$

Step 7: Introduce the duty cycle $D = T_{on} / T$ ($0 \leq D \leq 1$), the fraction of the period the switch spends ON:

$$V_o = D V_s$$

Why it is a Step-Down Chopper:

Because the duty cycle D is always a fractional value between 0 and 1, multiplying the source voltage V_s by D guarantees that the output voltage V_o is mathematically forced to be less than or equal to the input source voltage. Thus, it operates exclusively as a step-down chopper. V_o can be continuously varied from 0 V ($D = 0$) up to V_s ($D = 1$) simply by controlling the ON time of the single switch.

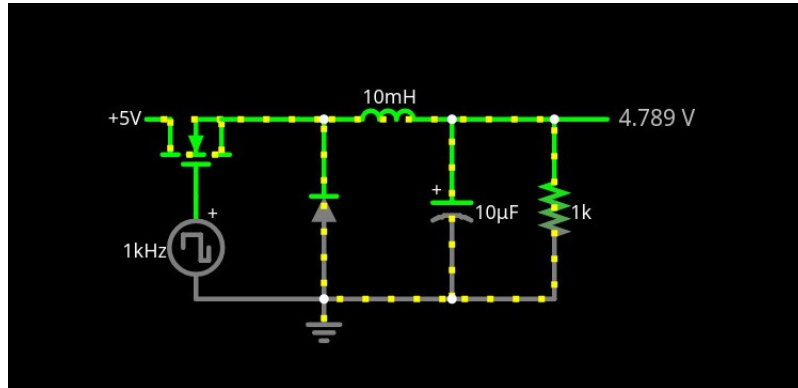


Figure 1(a): Falstad simulation circuit of a Buck Converter, structurally analogous to a Class A Chopper.

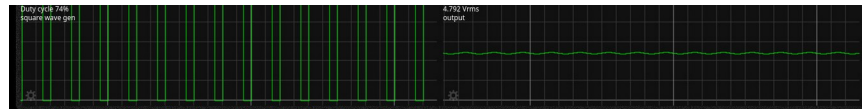


Figure 1(b): Simulated switching gate signal (left) and resulting output ripple waveform (right) at ~74% duty cycle, output ≈ 4.79 V from a 5 V source.

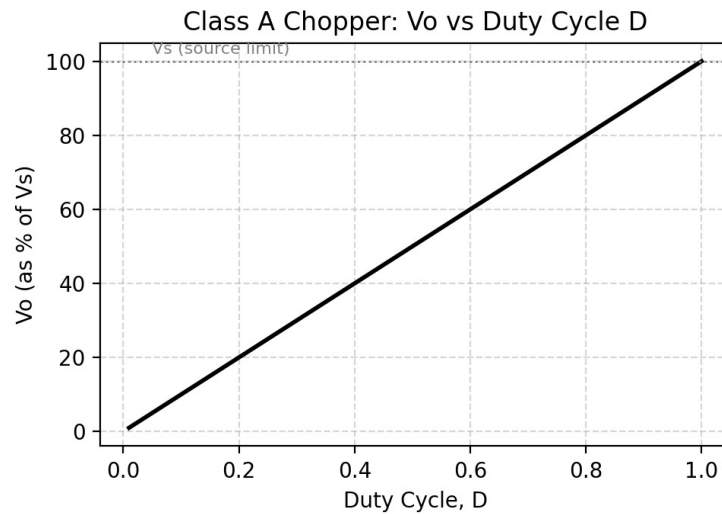


Figure 1(c): Output voltage V_o (as a percentage of V_s) plotted against duty cycle D — a straight line through the origin, confirming $V_o = D \cdot V_s$.

2. Class B Chopper (Second-Quadrant / Step-Up Chopper)

Circuit Description:

A Class B chopper operates in the second quadrant, where power flows from an active load — such as a spinning DC motor generating a back-EMF, E — back to the main DC source V_s . It is the chopper equivalent of a Boost converter, using an inductor, a single switch, and a diode arranged to step up a lower voltage E to charge the higher-voltage source V_s .

Step-by-Step Derivation:

Step 1: When the switch is ON (0 to T_{on}), the inductor is connected directly across the back-EMF source E , so the inductor voltage is $V_L = E$, and the inductor current ramps up, storing energy.

Step 2: When the switch is OFF (T_{on} to T), the inductor discharges its stored energy into V_s through a diode. Since the inductor now opposes the change in current, its voltage becomes $V_L = E - V_s$.

Step 3: For steady-state (periodic) operation, the net change in inductor current over one full period must be zero. This is the volt-second balance principle: the average voltage across the inductor over one period must equal zero.

$$\int_0^{T_{on}} E \, dt + \int_{T_{on}}^T (E - V_s) \, dt = 0$$

Step 4: Evaluate each integral over its respective interval:

$$E \cdot T_{on} + (E - V_s) \cdot T_{off} = 0$$

Step 5: Expand the bracket and collect the E terms:

$$E(T_{on} + T_{off}) = V_s \cdot T_{off}$$

Step 6: Since $T_{on} + T_{off} = T$, and $T_{off} = T - T_{on}$:

$$E \cdot T = V_s (T - T_{on})$$

Step 7: Divide both sides by T and use $D = T_{on} / T$, so that $(T - T_{on})/T = (1 - D)$:

$$V_s = E / (1 - D)$$

Why it is a Step-Up Chopper:

From the perspective of the power source (which is now the active load E), the voltage is being boosted to push current back into the main DC supply V_s . Since $(1 - D)$ is a fraction strictly less than 1 for any $D > 0$, dividing E by this fraction proves that the effective voltage presented to V_s is always greater than E — i.e., the generated back-EMF E is stepped up to match and overcome the higher supply voltage V_s . As D approaches 1, the theoretical step-up ratio grows without bound.

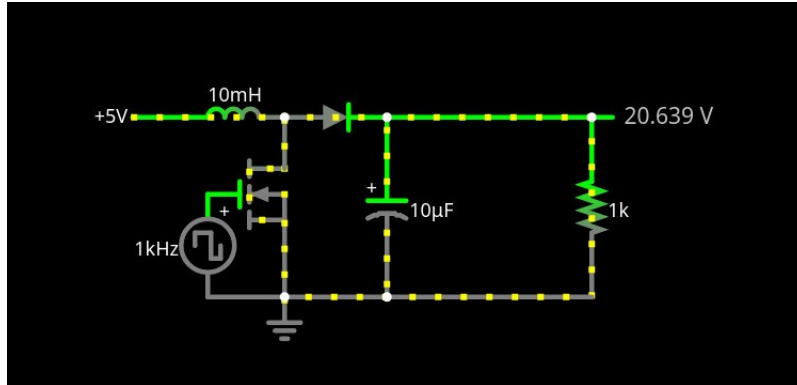


Figure 2(a): Falstad simulation circuit of a Boost Converter, demonstrating the step-up nature of the Class B Chopper topology.

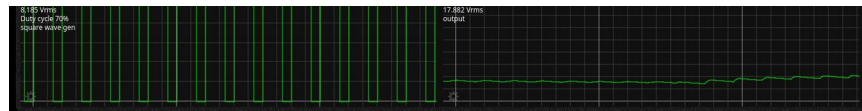


Figure 2(b): Switching signal (left, ~70% duty cycle) and boosted output waveform (right), stepping a 5 V source up to ≈ 20.6 V.

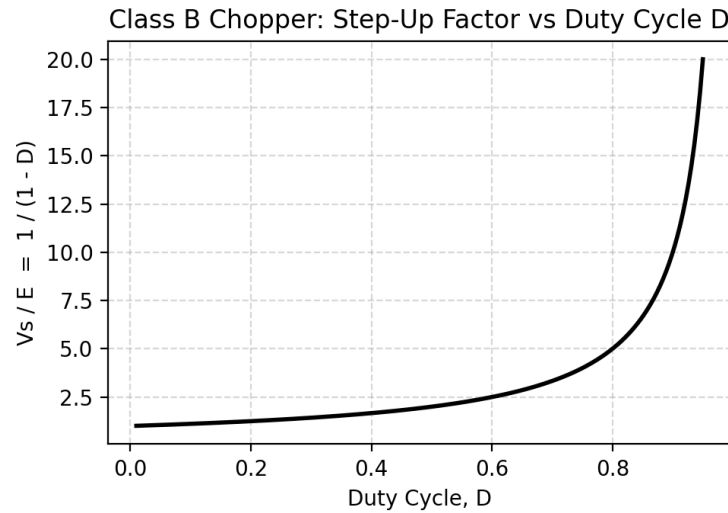


Figure 2(c): Step-up factor $V_s/E = 1/(1 - D)$ plotted against duty cycle D — the ratio rises sharply and approaches infinity as $D \rightarrow 1$.

3. Class C Chopper (Two-Quadrant Type A)

Circuit Description:

A Class C chopper is formed by combining a Class A (step-down) chopper and a Class B (step-up) chopper in parallel across the same load, using two switches (S1, S2) and two diodes (D1, D2), as shown in Figure 3(a). It can operate in the first quadrant (forward motoring: V_o and I_o both positive) or the second quadrant (forward braking / regeneration: V_o positive, I_o negative). In both quadrants V_o is restricted to a single (positive) polarity — only the direction of current, and hence the direction of power flow, reverses.

Step-by-Step Derivation — Motoring Mode (S1, D2 active):

Step 1: When switch S1 is ON (S2 and both diodes non-conducting), the load is connected directly to the source, exactly as in the Class A chopper: $v(t) = V_s$.

Step 2: When S1 is OFF, the load current freewheels through diode D2, making the instantaneous output voltage zero: $v(t) = 0$ — again identical to the Class A operating mode.

Step 3: Averaging $v(t)$ over one switching period T , exactly as performed for the Class A chopper:

$$V_o = (1/T) \left(\int_0^{T_{on}} V_s dt + \int_{T_{on}}^T 0 dt \right) = (V_s \cdot T_{on}) / T$$

Step 4: Substituting the duty cycle $D = T_{on} / T$:

$$V_o = D V_s$$

Step-by-Step Derivation — Regenerative Braking Mode (S2, D1 active):

Step 5: When switch S2 is ON, the inductive/motor branch (back-EMF E) stores energy, exactly as the switch-ON interval of the Class B chopper.

Step 6: When S2 is OFF, diode D1 conducts, discharging the stored energy back into the source V_s , exactly as the switch-OFF interval of the Class B chopper.

Step 7: Applying the same volt-second balance derived for Class B directly gives the relation between source and load back-EMF:

$$V_s = E / (1 - D)$$

Why it is Step-Down in One Mode and Step-Up in the Other:

In motoring (first-quadrant) operation, the Class C chopper behaves exactly like the Class A chopper: since $0 \leq D \leq 1$, $V_o = D V_s$ is always less than or equal to V_s , so it functions as a step-down chopper. In regenerative braking (second-quadrant) operation, it behaves exactly like the Class B chopper: the factor $1/(1 - D)$ is always greater than or equal to 1, so the load back-EMF E is stepped up to overcome V_s and return power to the source. The key distinguishing feature of the Class C chopper is that, unlike Class D or E, the output voltage polarity never reverses.

Note on Figure 3: The original Falstad file supplied for this assignment did not include a saved simulation screenshot for the Class C topology (only Class A, B, and the Appendix converters

were captured). The circuit diagram below has therefore been redrawn from the standard two-quadrant Type-A chopper configuration to illustrate the switch/diode arrangement referred to in the derivation above.

Class C Chopper — Two-Quadrant (Type A) Configuration

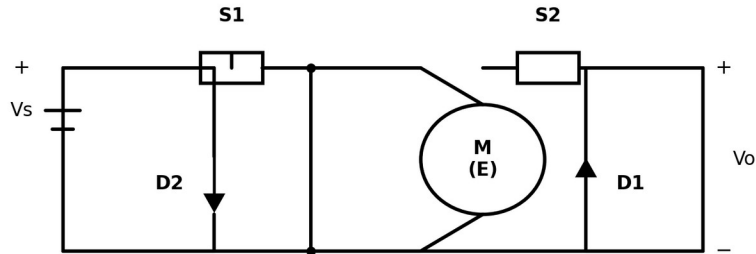


Figure 3(a): Reconstructed schematic of the Class C (two-quadrant Type A) chopper, showing switches S1/S2 and diodes D1/D2 arranged around the motor load M (back-EMF E).

Class C Chopper: Two-Quadrant Operation

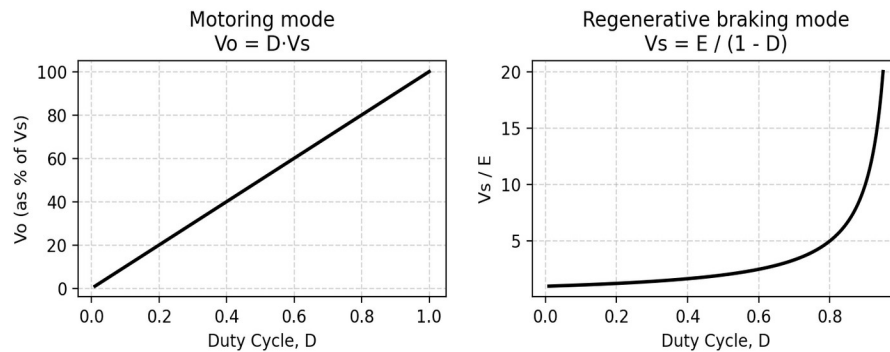


Figure 3(b): Output-voltage behaviour of the Class C chopper in its two operating modes — motoring (left, $V_o = D \cdot V_s$) and regenerative braking (right, $V_s = E/(1 - D)$).

4. Class D Chopper (Two-Quadrant Type B)

Circuit Description:

In a Class D chopper, two diagonal pairs of switches (forming a half of an H-bridge) connect the load to $+V_s$ when the first pair is ON, and to $-V_s$ when the second pair is ON (current freewheels back through the feedback diodes during the complementary interval). This allows the average output voltage to take either polarity while the current remains unidirectional — a two-quadrant (Type B) converter.

Step-by-Step Derivation:

Step 1: During T_{on} , the diagonal switch pair connects the load to the positive rail: $v(t) = +V_s$.

Step 2: During $T_{off} = T - T_{on}$, the complementary switch pair (or feedback diodes) connects the load to the negative rail: $v(t) = -V_s$.

Step 3: Average the instantaneous voltage over one period, splitting the integral at T_{on} :

$$V_o = (1/T) \left(\int_0^{T_{on}} V_s dt + \int_{T_{on}}^T (-V_s) dt \right)$$

Step 4: Evaluate both integrals:

$$V_o = (1/T) [V_s \cdot T_{on} - V_s \cdot (T - T_{on})]$$

Step 5: Factor out V_s and simplify the bracket in terms of T_{on}/T :

$$V_o = V_s (2T_{on}/T - 1)$$

Step 6: Substitute $D = T_{on}/T$:

$$V_o = V_s (2D - 1)$$

Why it is a Step-Down Chopper (with polarity reversal):

If $D > 0.5$, V_o is positive. If $D < 0.5$, V_o is negative. In all cases, the absolute maximum magnitude of the output voltage $|V_o|$ can never exceed V_s (it only reaches $\pm V_s$ at the extreme duty-cycle limits $D = 1$ or $D = 0$). Therefore, it remains fundamentally a step-down chopper, but with the unique added ability to produce a continuously variable negative average voltage as well as a positive one — essential for four-quadrant motor speed and torque control.

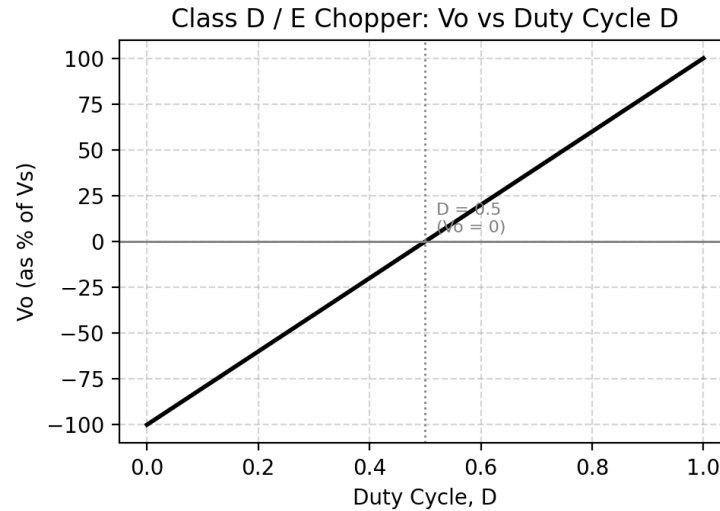


Figure 4: Output voltage V_o (as % of V_s) versus duty cycle D for the Class D / E chopper — $V_o = V_s(2D - 1)$. V_o crosses zero at $D = 0.5$ and is bounded within $\pm V_s$.

5. Class E Chopper (Four-Quadrant)

Circuit Description:

A Class E chopper utilizes a full H-bridge topology (four switches and four diodes) to operate in all four quadrants (forward/reverse motoring and forward/reverse braking).

Step-by-Step Derivation:

Step 1: Under a bipolar PWM switching strategy, exactly one diagonal switch pair is ON at any instant, and the two diagonal pairs alternate — this is identical in principle to the Class D switching pattern.

Step 2: The same volt-second averaging performed for the Class D chopper therefore applies directly, giving:

$$V_o = V_s (2D - 1)$$

Step 3: Under a unipolar PWM switching strategy instead, only one pair of switches is chopped at a time while the other pair is held fixed, giving a simpler relationship in each active region:

$$V_o = D V_s \quad (\text{forward active region}) \quad V_o = -D V_s \quad (\text{reverse active region})$$

Why it is a Step-Down Chopper:

Regardless of which quadrant the motor is operating in, the H-bridge is fundamentally a buck-derived topology — it can only switch the load between $+V_s$, 0, and $-V_s$ (or between $+V_s$ and $-V_s$ for bipolar PWM); it never contains an inductive boost stage referenced to V_s . The output voltage magnitude is therefore always a fractional portion of the DC link voltage, strictly enforcing step-down operation in every quadrant.

Appendix: Advanced Chopper / Converter Topologies

To further illustrate variations of DC-DC conversion and inverted voltage mapping, the following simulations demonstrate complex energy-transfer topologies derived from the same basic chopper (switch + energy-storage element) principles used above.

Ćuk Converter:

The Ćuk converter uses a capacitor, rather than only an inductor, as the primary energy-transfer element between input and output, giving a regulated, polarity-inverted output with continuous (non-pulsating) input and output currents — a useful property not shared by the Buck or Boost topologies above.

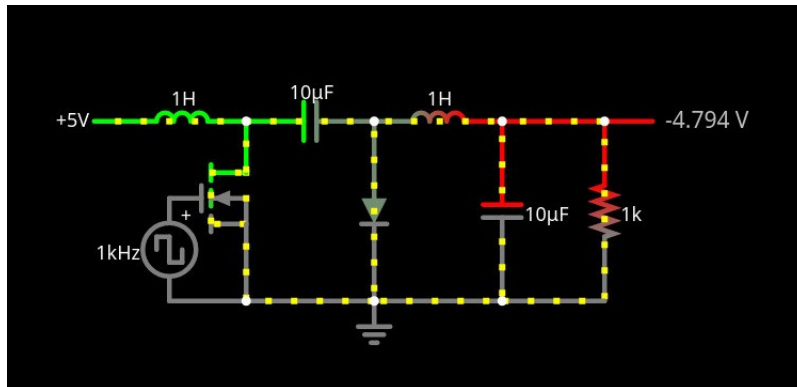


Figure 5(a): Falstad simulation circuit of a Ćuk Converter.

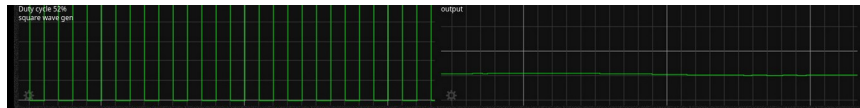


Figure 5(b): Switching waveform (left) and inverted, regulated output (right), stepping a 5 V input down to approximately -4.79 V.

SEPIC (Single-Ended Primary-Inductor Converter):

The SEPIC topology allows the output voltage to be greater than, less than, or equal to the input voltage, while maintaining a non-inverted (positive) output polarity — combining the flexibility of the Ćuk converter with the same-sign output of a Buck-Boost converter.

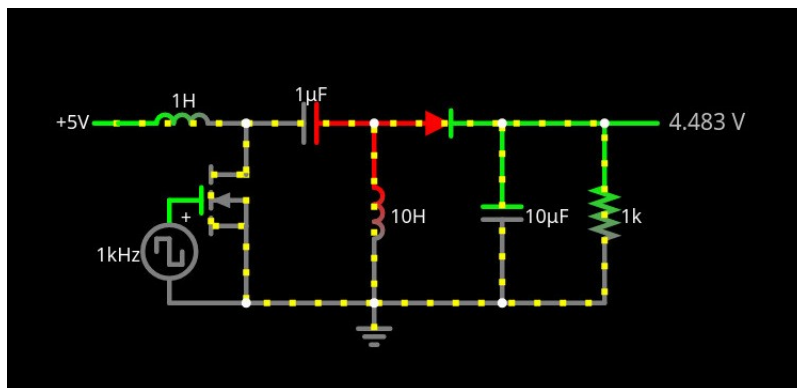
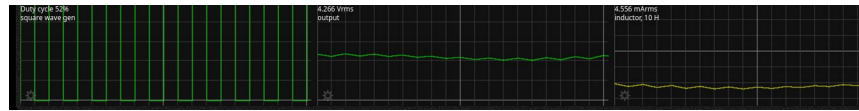


Figure 6(a): Falstad simulation circuit of a SEPIC Converter.

Figure 6(b): Switching waveform (left) and regulated, non-inverted output (right), ≈ 4.48 V from a 5 V source.

Conclusion

This assignment derived the average output-voltage equation for all five standard chopper classes from first principles, using time-domain averaging (Classes A, C-motoring, D, E) and inductor volt-second balance (Classes B, C-regeneration). The results are summarised below:

Class	Quadrant(s)	Output Voltage Equation	Nature
A	1st (motoring)	$V_o = D V_s$	Step-down
B	2nd (regen)	$V_s = E / (1 - D)$	Step-up
C	1st & 2nd	$V_o = D V_s / V_s = E / (1 - D)$	Step-down & Step-up (same polarity)
D	1st & 2nd (Type B)	$V_o = V_s (2D - 1)$	Step-down, polarity-reversing
E	All four	$V_o = V_s (2D - 1)$ [bipolar]	Step-down, all quadrants

A clear progression emerges across the five classes. Class A and Class B are the elementary building blocks — pure step-down and pure step-up converters, respectively, each operating in a single quadrant. Class C combines both of these elementary behaviours to allow bidirectional power flow (motoring and regenerative braking) while keeping the output voltage unipolar. Class D goes a step further, allowing the output voltage itself to reverse polarity (two-quadrant Type B), at the cost of always remaining a step-down converter with $|V_o| \leq V_s$. Class E generalises Class D to a full four-quadrant H-bridge, so that a single chopper can drive a motor forward or backward and brake it in either direction, while its voltage equation, under bipolar PWM, remains identical in form to that of Class D.

The Falstad simulations confirm these derivations experimentally: the Buck converter (Class A analogue) reduced a 5 V source to about 4.79 V at 74% duty cycle; the Boost converter (Class B analogue) raised the same 5 V source to about 20.6 V at 70% duty cycle; the Ćuk converter produced an inverted, regulated output of about -4.79 V; and the SEPIC converter produced a non-inverted, regulated output of about 4.48 V — all consistent with the governing equations derived above.

Discussion

- The step-down (Class A) and step-up (Class B) relations both reduce to the same physical statement — the average voltage across an inductor must be zero over one switching period in steady state — even though Class A was derived without invoking an inductor

explicitly; in practice a real Class A buck stage always includes an output inductor, and the $D \cdot V_s$ result can equally be obtained from volt-second balance on that inductor.

- The $(1 - D)$ term that appears in every step-up relation (Class B, and the regenerative mode of Class C) is the fundamental reason boost-type converters cannot achieve a stable, finite output as D approaches 1 — in a practical circuit, parasitic resistances cause the real gain to peak and then fall, unlike the idealised curve plotted in Figure 2(c).
- Classes D and E trade a simpler, purely positive output (as in Class A) for bipolar output capability, which is essential wherever a load — most commonly a DC motor — must be driven and braked in both directions; this is achieved at no cost to component count efficiency beyond needing a full or half H-bridge instead of a single switch.
- The Ćuk and SEPIC converters in the Appendix demonstrate that the same duty-cycle-controlled switching principle extends naturally beyond the five classical chopper classes into converters using capacitive energy transfer, offering additional benefits (inverted output, continuous currents, or flexible step-up/step-down operation) not available from a simple Class A or B chopper.

Software Used

All chopper and converter circuits in this assignment were simulated using the Falstad Circuit Simulator (an open-source, browser-based SPICE-like simulator by Paul Falstad), from which the circuit schematics and oscilloscope waveform captures reproduced in Figures 1, 2, 5, and 6 were taken. The duty-cycle response graphs (Figures 1(c), 2(c), 3(b), and 4) were generated using Python 3 with the Matplotlib plotting library. The Class C schematic (Figure 3(a)) was reconstructed using the same tool, as no saved simulation was available for that topology. The final document was typeset programmatically in Microsoft Word (.docx) format using the docx (docx-js) library and exported to PDF.

References

- [1] Falstad, P., "Circuit Simulator Applet," [Online Software]. Available: <https://www.falstad.com/circuit/>
- [2] Rashid, M. H., Power Electronics: Circuits, Devices, and Applications, 4th ed., Pearson, 2014 — Chapter on DC Choppers (Classes A–E).
- [3] Mohan, N., Undeland, T. M., and Robbins, W. P., Power Electronics: Converters, Applications, and Design, 3rd ed., Wiley, 2003.
- [4] Hunter, J. D., "Matplotlib: A 2D Graphics Environment," Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.
- [5] Course lecture notes / assignment brief on DC Chopper (Class A–E) output voltage derivations, as provided by the instructor.